

Colon Cancer Prediction On Different Magnified Colon Biopsy Images

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Abstract—Colon Cancer detection is an important task for the histopathologist as they have to analyze morphology of the images at different magnifications thereby leading to intra and inter observer variability. Thus an automation is needed to detect the colon cancer for all the magnifications. This paper focus on the colon cancer detection on the colon biopsy images at different magnifications. First the images are converted to its HSV color space and further processing are performed on the saturation component. Features are the coefficients obtained from Dual-tree and double-density 2-D wavelet transform. In the last phase classification is performed with the Random Forest classifier. The proposed system was evaluated on colon biopsy images collected from Aster Medcity at different magnifications 10X, 20X, 40X and combined magnifications where an accuracy of 85.4% is obtained for the combined magnifications.

Index Terms—Colon Cancer, biopsy images, magnification, feature, classification

I. INTRODUCTION

Colon cancer is the cancer which occurs in the colon of the large intestine and its one of the leading cancer that occurs today. Colon cancer stands second and third in the death cause and incident case respectively [1]. Colon cancer is confirmed through the analysis of colon biopsies that are collected during colonoscopy or surgery by the histopathologist. The pathologist analyze the morphology or shape of the biopsy images through the microscope and detects cancer. While doing so, the observations made by one pathologist may be different from other and hence may lead to inter and inter observer variability [2], [3]. In order to have better visibility the pathologist observe the slides at different magnifications and hence cannot stick on to single magnifications. Hence an automation is necessary to detect the colon cancer on different magnified images so as to assist the pathologist.

II. LITERATURE REVIEW

In these days medical imaging helps to identify the diseases in various part of the body. Various machine learning tools are used in medical image processing for the pattern recognition and draws conclusions for the medical images [4], [5]. Various researches have been carried out for the detection of colon cancer from the biopsy images. A detailed survey of this are summarized by Saima et al. [6] where simple and object

oriented features proved to be best for the detection of cancer. With the texture feature such as entropy, contrast, correlation, angular second moment, dissimilarity and inverse difference moment that are extracted from the GLCM matrix, Esgiar et al. [7] detects colon cancer Linear discriminant analysis and K-nearest Neighbor from 40X magnified image and achieved an accuracy of 90.2%. E. Ozdemir et al. [8] by generating multiple sequences of image with Markov process detects colon cancer on 20X magnified image with an accuracy of 90.32%. With 10X magnified 82 normal and 92 malignant colon images Saima Rathore et.al [9] detects colon cancer by extracting minimal hybrid feature by minimum Redundancy Maximum Relevance (mRMR) on color components based statistical moments(CCSM), Haralick features and HOG. Then classified with ensemble classifier of voting is where attained an performance accuracy of 98.85%. With the same images Saima Rathore et.al [10] detects cancer with the hybrid feature consisting of texture, morphological, elliptic Fourier descriptors(EFDs), and scale invariant feature transform(SIFT) of 244 in length and obtained 98.07%. Colon cancer detection and grading was done by Saima Rathore et.al [12] on 10X magnified images based on the lumen characteristics and attained detection accuracy of 95.4% and grading accuracy of 93.47%. With different magnification 4X, 5X, 10X and 40X Saima Rathore et.al [11] identifies cancer by finding three novel features namely, semi-major axis, direction, and area occurrence of the ellipse from the segmented image. Genetic algorithm is used to optimize the parameters for each of the magnification and classified using RBF classifier where an accuracy of 92.33% was obtained. In the initial experiments of [13], [14] colon cancer was detected on the different magnified images using texture and morphological features where an accuracy of 91.3% was obtained with 10X magnification using the MLP classifier.

Most the colon cancer detection techniques are done on single magnification factors as the features are dependent on the magnification and may vary for the other magnifications. The morphological features purely depends on the shape and structure and thus is highly depend on the magnification.

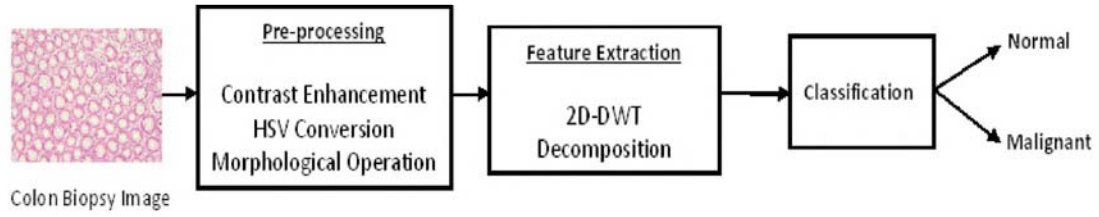


Fig. 1. Framework Of the Proposed System

Hence features used in one magnification need not work properly for the other magnifications. If we consider the spatial domain, the morphological features play in role that is dependent on the magnification. Thus an automated system is essential to detect colon cancer that works for all the magnifications.

This paper focus on the detection of colon cancer which could work in all the magnifications. First the RGB colon biopsy images are converted to the HSV color space after the contrast enhancement. From the HSV color space, the saturation component is taken and morphological operation is performed on it and converted back to RGB color space. In the next step, Dual-tree and double-density 2-D wavelet transform is performed on gray scale converted images and the coefficients are taken as the features for classification. In the last phase, the images are classified as normal or malignant with the Random Forest Classifier.

III. PROPOSED METHODOLOGY

The framework of the proposed methodology which work across all the magnifications consist of 3 stages namely (1) Pre-processing (2) Feature Extraction (3) Classification. Figure.1 shows the framework of the proposed system.

A. Pre-processing

The colon biopsy images obtained after the acquisition cannot be used directly for the image processing. It has to be made good enough and cleaned in such a way that it could be used well in the subsequent processing. First the RGB colon biopsy images are re-sized to 256 x 256 and then the contrast enhancement is performed on the three color bands with adaptive histogram equalization. Then the images are converted to HSV color space as this color space are much consistent with human color perception and highly stable under illumination changes. From this only the saturation component is taken and then morphological operations are performed. The preprocessed image is shown in Figure.2. Then morphological opening is done followed by closing. The final image is then converted back to RGB color space and its converted to its gray scale.

B. Feature Extraction

The colon biopsy images needs to be quantified such that classification has to be performed on that. The features should be chosen in such a way that it would uniformly work with all the magnifications. For that the spatial domain features

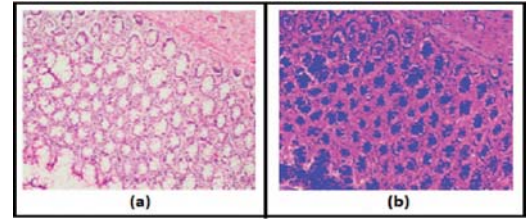


Fig. 2. Pre-processing (a) Raw Image (b) Pre-processed Image

could not be taken. Hence the image should be transformed to the frequency domain. Thus 2D-Dual-tree and double-density 2-D wavelet transform is performed on the pre-processed image with the Haar mother wavelet. So that it could perform in all the magnifications irrespective of the morphological features. With the Dual-tree and double-density 2-D wavelet transform. The decomposition is done for three levels and the final obtained coefficients are taken as the features. The coefficients obtained for the normal images would be higher in value whereas lower for the malignant images as shown in Figure.3. The features are computed for the colon images taken at different magnifications.

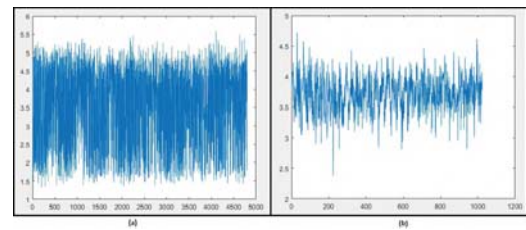


Fig. 3. Dual-tree and double-density 2-D wavelet transform coefficients (a) Normal Image (b) Malignant Image

C. Classification

The features extracted after the Dual-tree and double-density 2-D wavelet transform is fed into the classifier for classification. Random Forest classifier is used for the binary classification and thus classifies into normal and malignant ones.

IV. PERFORMANCE EVALUATION MEASURES

The performance of the proposed model needs to be evaluated and its done how well the classifier classifies the

image samples. The standard performance evaluation measures include Accuracy, F-Score, Area Under Curve(AUC). These measures are obtained from the confusion matrix created. The positive and negative sample is considered as the normal and malignant images respectively. True positives(TN) and True negatives(TN) corresponds to the correctly classified normal and negative images respectively whereas False positives(FP) and False negatives(FN) indicates the misclassified normal and malignant images respectively. Following gives the brief description of the measures used.

Accuracy gives the overall performance of the classifier and its the number of the correctly classified images in the total population N and is given by

$$Accuracy = \frac{TP + TN}{N} \times 100 \quad (1)$$

F-Score gives the precision of the classifier and its the weighted average of recall and precision. Its given by

$$F - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (2)$$

where $Precision = \frac{TP}{TP+FP}$ and $Recall = \frac{TP}{TP+FN}$

The F-Score values ranges from 0 to 1 where near to 0 means worst and near to 1 means good.

Area under the Curve(AUC) also quantifies how well a classifier is. Its values ranges from 0 to 1, where close to means good, else worst.

Matthews correlation coefficient(MCC) value is important for binary classification where it gives the measure ranges from -1 to +1. If the classifier classifies a correct label, then its +1 and -1 if it predicts a wrong label. If its 0, then its predictions are random. Its calculated as in Equation.3

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FN)(TP+FP)(TN+FN)(TN+FP)}} \quad (3)$$

V. DATA AND EXPERIMENTAL SETUP

The dataset consist of colon biopsy images collected from Aster Medcity, Kochi, India at different magnifications 10X, 20X and 40X. The images were captured with Nikon eclipse Ci with NIS element viewer microscope from H&E stained colon samples that are 5-6 μ m thick tissue section. The dataset were prepared and ground truth labeling for the images were done by Dr. Sarah Kuruvila, Senior consultant and Dr. Shahin Hameed, Specialist, Pathology Department, Aster Medcity, Kochi, India. The details of the dataset is given in Table.I The features are extracted using Matlab 2017b and the classi-

TABLE I
COLON BIOPSY IMAGE DATASET

Dataset	Magnification	Normal	Malignant
Aster Medcity	10 X	70	75
	20 X	70	75
	40 X	70	75

fication is done with Weka. In the preprocessing phase, for the morphological closing operation a disc with 3 size is

used whereas for the opening disc with size 7 was used. For the Dual-tree and double-density 2-D wavelet transform, Haar wavelet is used as the mother wavelet. The features are tabulated for the images captured at different magnification and to analyze for the magnification independency, the features were extracted for all the magnification and were taken together for the classification. K-fold cross validation with K=10 is done and then classified with the Random Forest Classifier.

VI. RESULTS AND DISCUSSION

Table.II shows accuracy, F-score, AUC for and MCC for the individual magnifications and combination of all magnifications.

TABLE II
PERFORMANCE EVALUATION MEASURES

Magnification	Accuracy	F-Score	AUC	MCC
10 X	85.49	0.855	0.929	0.706
20 X	87.69	0.875	0.944	0.756
40 X	86.8	0.867	0.947	0.735
Combined	85.3	0.852	0.935	0.695

When accuracy is analyzed, with all magnified images its 85.3% which is almost the same as that of the individual magnifications. 20X magnification better performs in the proposed model with an accuracy of 87.69% followed by 40X and 10X. But then also when the combined magnification is taken, the proposed system has got good accuracy even though with the large image set as the images in all the magnifications are being added. This is because, the number of images is more as its the sum of the images in the individual magnifications. When F-Score and AUC is considered, for the combined magnifications its giving a value of 0.852 and 0.935, which indicates the proposed system model is good and almost near to the individual magnifications. To analyze the prediction capability of the classification, MCC is found to be 0.695 for the combined magnification which is near to 1. Hence this also indicates the proposed model is good for all the magnifications. Thus the proposed system works well across all the magnifications. The direct comparison with other colon detection techniques is not possible as no paper has handled with multiple magnifications.

VII. CONCLUSION

This paper focus on the colon cancer detection method that could work on the colon biopsy images of all magnifications taken together. During the pre-processing stage, the RGB images are converted to its HSV color space and only the saturation component is taken for the processing. In the feature extraction phase, Dual-tree and double-density 2-D wavelet transform and its coefficients is used for the classification. Later Random Forest classifier is used to classify where an accuracy of 85.3% is obtained for the combined magnification. There are many future possibilities where the work could be extended. First many other magnification independent features could be tried so that performance is improved. Second, colon

cancer grading with the magnification independent features could be tried.

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